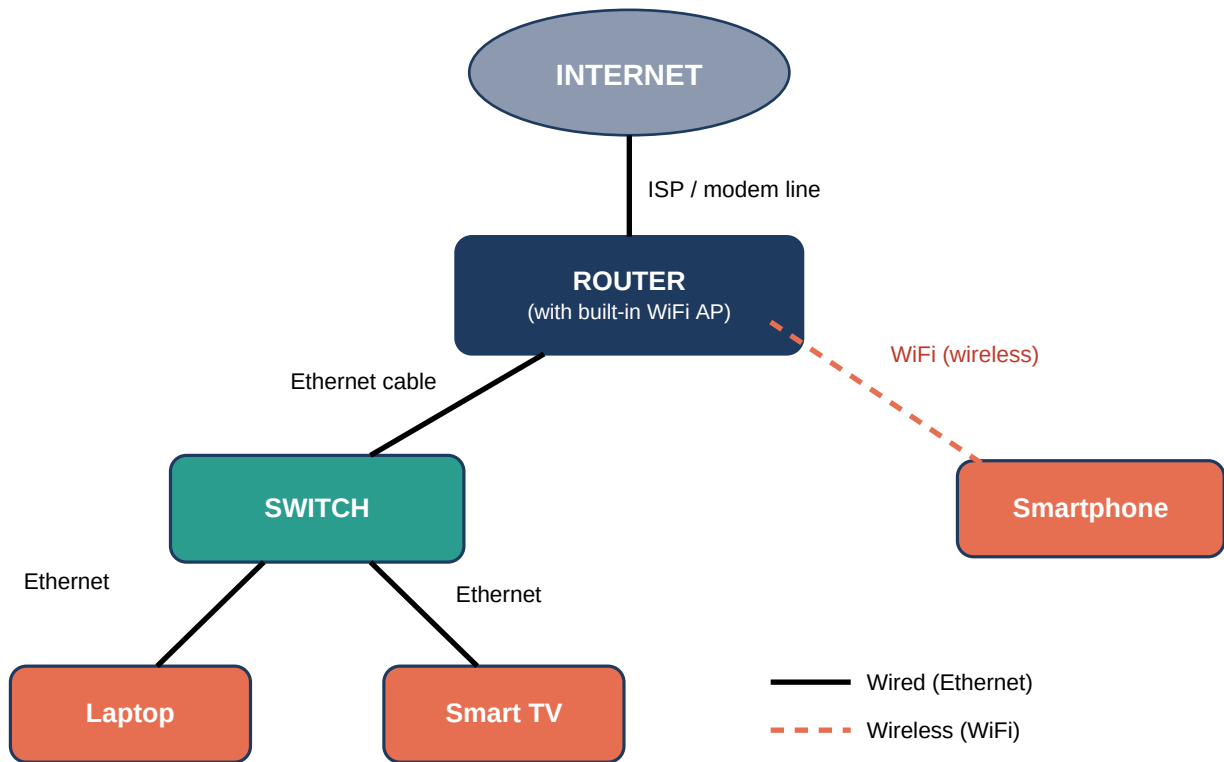


Network Devices

Questions and Answers

Q1. Simple Home Network Diagram

Draw and label a simple home network diagram showing a router, switch, and three devices, indicating how each device connects.



- **Router** connects the whole home to the Internet through the ISP line.
- **Switch** is plugged into the router with an Ethernet cable and gives extra wired ports.
- **Laptop** and **Smart TV** connect to the switch using Ethernet cables (wired, stable, fast).
- **Smartphone** connects wirelessly over WiFi to the router's built-in access point.

Q2. Comparison Table of Network Devices

List at least four network devices and describe one real-world use case for each.

Device	What it does	Real-world use case
Router	Connects your local network to the Internet and directs data packets between networks using IP addresses.	The router at home lets your phone and laptop reach the Internet for YouTube, Instagram and Zomato.
Switch	Connects wired devices in a LAN and sends data only to the device it is meant for (uses MAC addresses).	LAN gaming: several PCs at a gaming cafe connect to a switch for fast, low-lag multiplayer matches.
Hub	Basic device that copies incoming data to all connected ports. No filtering, so it is slower and less secure.	Old small offices or labs sharing a printer or files between a few PCs. Today it is mostly replaced by switches.
Access Point (AP)	Creates a WiFi signal so wireless devices can join the wired network.	College or mall WiFi: AP units on ceilings let students stream Netflix or join Google Meet on phones.

Q3. How Data Travels When Ordering on Zomato

Explain how data travels from your phone to the internet, mentioning the role of each device.

I open the Zomato app on my phone and tap **Place Order**. Here is the journey of that request:

- **Phone to Access Point:** The phone turns my order into data packets and sends them as WiFi radio signals to the access point.
- **Access Point to Switch:** The access point converts the wireless signal into wired data and passes it through an Ethernet cable to the switch.
- **Switch to Router:** The switch looks at the destination address and forwards the packets only to the router, not to every device.
- **Router to Internet:** The router sends the packets through the modem and ISP to the Internet and finds the best path to Zomato's servers.
- **Zomato server and back:** The server confirms the order and sends a reply (order placed, delivery time). It travels back the same way: router, switch, access point, my phone.

In short: **Phone -> Access Point -> Switch -> Router -> Internet -> Zomato server**, and the confirmation returns along the same path in a fraction of a second.

Q4. Connecting a Far-Away Gaming PC (Ethernet Only)

You have a WiFi router, but your gaming PC has only an Ethernet port and is far from the router. Suggest two devices.

Option 1: Powerline Adapter Kit

This kit has two adapters. Plug the first into a wall socket near the router and connect it to the router with an Ethernet cable. Plug the second into a socket near the gaming PC and connect it to the PC with an Ethernet cable. The adapters send the network data through the home's existing electrical wiring, so no long cable is needed. It gives a stable wired connection with low lag, though speed depends on the quality of the house wiring.

Option 2: WiFi Extender / Wireless Bridge with an Ethernet Port

Place the extender midway, where it still gets a strong signal from the router. It receives the router's WiFi and re-broadcasts it. Plug the gaming PC into the extender's Ethernet port (or use a WiFi-to-Ethernet adapter), and the PC gets internet over the wireless link even though the PC itself has no WiFi. It is easy to set up with no wiring, but speed and ping can be slightly worse than a wired link.

Note on switches

A switch alone does not solve distance. It only adds more Ethernet ports, so it is useful only if a long Ethernet cable can run from the router to the PC area, or to connect several devices there.

Best choice: Powerline adapters for gaming (more stable, lower lag); a WiFi extender if the wiring is old or you want zero cables.